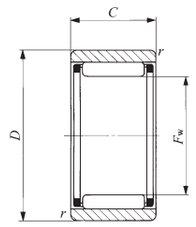
NEEDLE ROLLER BEARINGS WITH SEPARABLE CAGE

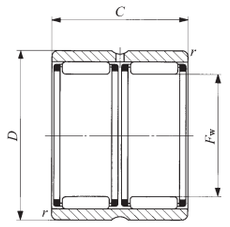
Without Inner Ring



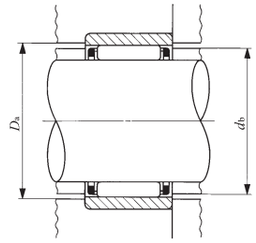


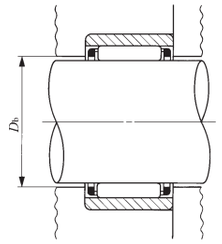


RNAF



RNAFW





Shaft dia. 45 — 100mm

Shaft dia. mm	Identification number	Mass (Ref.) g	Boundary dimensions mm F_w	D	C	$r_{s\,min}$ (1)	Standard mounting dimensions mm d_b	D_a Max.	D_b	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
45	RNAF 455517	83.5	45	55	17	0.3	48.5	53	45.8	23 300	47 100	11 000
	RNAFW 455534	167	45	55	34	0.3	48.5	53	45.8	39 900	94 200	11 000
	RNAF 456220	184	45	62	20	0.3	50.9	60	45.8	33 200	53 300	11 000
	RNAFW 456240	370	45	62	40	0.3	50.9	60	45.8	56 900	107 000	11 000
50	RNAF 506220	138	50	62	20	0.3	54.2	60	50.8	27 100	59 300	10 000
	RNAFW 506240	275	50	62	40	0.3	54.2	60	50.8	46 400	119 000	10 000
	RNAF 506520	170	50	65	20	0.3	55.2	63	50.8	35 900	61 100	10 000
	RNAFW 506540	340	50	65	40	0.6	55.2	61	50.8	61 500	122 000	10 000
55	RNAF 556820	167	55	68	20	0.3	59.5	66	55.8	28 600	66 000	9 000
	RNAFW 556840	335	55	68	40	0.3	59.5	66	55.8	49 000	132 000	9 000
	RNAF 557220	220	55	72	20	1	60.9	67	55.8	37 400	66 400	9 000
	RNAFW 557240	440	55	72	40	1	60.9	67	55.8	64 100	133 000	9 000
60	RNAF 607820	255	60	78	20	1	66.3	73	60.8	38 900	71 700	8 500
	RNAFW 607840	510	60	78	40	1	66.3	73	60.8	66 700	143 000	8 500
65	RNAF 658530	470	65	85	30	1.5	72	77	66	59 300	127 000	7 500
	RNAFW 658560	945	65	85	60	1.5	72	77	66	102 000	255 000	7 500
70	RNAF 709030	500	70	90	30	1.5	77	82	71	61 200	136 000	7 000
	RNAFW 709060	1 000	70	90	60	1.5	77	82	71	105 000	272 000	7 000
75	RNAF 759530	530	75	95	30	1.5	82	87	76	63 100	144 000	6 500
	RNAFW 759560	1 060	75	95	60	1.5	82	87	76	108 000	289 000	6 500
80	RNAF 8010030	560	80	100	30	1.5	87	92	81	65 000	153 000	6 000
	RNAFW 8010060	1 120	80	100	60	1.5	87	92	81	111 000	306 000	6 000
85	RNAF 8510530	590	85	105	30	1.5	92	97	86	66 600	161 000	6 000
90	RNAF 9011030	625	90	110	30	1.5	97	102	91	69 600	174 000	5 500
95	RNAF 9511530	655	95	115	30	1.5	102	107	96	70 900	182 000	5 500
100	RNAF 10012030	685	100	120	30	1.5	107	112	101	72 500	191 000	4 500

Notes(1) Minimum allowable value of chamfer dimension *r*
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 50% of this value is allowable.
Remarks1. RNAF has no oil hole. RNAFW is provided with an oil groove and an oil hole on the outer ring.
2. No grease is prepacked. Perform proper lubrication.

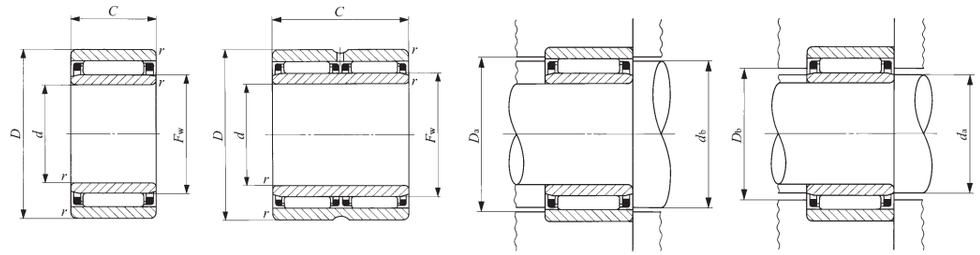
1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

NEEDLE ROLLER BEARINGS WITH SEPARABLE CAGE

With Inner Ring







NAF NAFW

Shaft dia. 6 — 25mm

Shaft dia. mm	Identification number	Mass (Ref.) g	Boundary dimensions mm d	D	C	$r_{s\,min}$ (1)	F_w	Standard mounting dimensions mm d_b	D_a Max.	d_a Min.	d_a Max.	D_b	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm	Assembled inner ring
6	NAF 61710	13.5	6	17	10	0.3	10	12.4	15	8	9.7	10.4	4 160	4 550	50 000	LRT 61010
7	NAF 72012	22.5	7	20	12	0.3	10	13.5	18	9	9.7	10.4	5 940	6 000	50 000	LRT 71012-1
9	NAF 92212	24	9	22	12	0.3	12	15.5	20	11	11.5	12.4	9 030	8 460	40 000	LRT 91212
10	NAF 102213	26	10	22	13	0.3	14	17.6	20	12	13	14.6	7 860	9 410	35 000	LRT 101413
	NAFW 102220	40	10	22	20	0.3	14	17.6	20	12	13	14.6	10 800	14 200	35 000	LRT 101420
	NAF 102612	36	10	26	12	0.3	14	19.4	24	12	13	14.6	9 790	9 680	35 000	LRT 101412
12	NAF 122413	29.5	12	24	13	0.3	16	19.6	22	14	15	16.6	8 620	11 000	30 000	LRT 121613
	NAFW 122420	45.5	12	24	20	0.3	16	19.6	22	14	15	16.6	11 900	16 700	30 000	LRT 121620
	NAF 122812	40	12	28	12	0.3	16	21.4	26	14	15	16.6	10 500	10 900	30 000	LRT 121612
15	NAF 152813	38.5	15	28	13	0.3	20	23.6	26	17	19	20.6	9 590	13 500	25 000	LRT 152013
	NAFW 152826	77.5	15	28	26	0.3	20	23.6	26	17	19	20.6	16 400	27 100	25 000	LRT 152026
	NAF 153212	50.5	15	32	12	0.3	20	25.4	30	17	19	20.6	12 400	14 300	25 000	LRT 152012
17	NAF 173013	42.5	17	30	13	0.3	22	25.6	28	19	21	22.6	10 200	15 200	25 000	LRT 172213
	NAFW 173026	84.5	17	30	26	0.3	22	25.6	28	19	21	22.6	17 500	30 300	25 000	LRT 172226
	NAF 173516	77.5	17	35	16	0.3	22	27.8	33	19	21	22.6	17 600	20 900	25 000	LRT 172216
	NAFW 173532	155	17	35	32	0.3	22	27.8	33	19	21	22.6	30 200	41 800	25 000	LRT 172232
20	NAF 203517	74	20	35	17	0.3	25	29.5	33	22	24	25.6	17 300	26 600	20 000	LRT 202517
	NAFW 203526	114	20	35	26	0.3	25	29.5	33	22	24	25.6	22 400	37 200	20 000	LRT 202526
	NAF 203716	79	20	37	16	0.3	25	30.4	35	22	24	25.6	19 400	24 500	20 000	LRT 202516
	NAFW 203732	158	20	37	32	0.3	25	30.4	35	22	24	25.6	33 200	49 000	20 000	LRT 202532
25	NAF 254017	87.5	25	40	17	0.3	30	34.5	38	27	29	30.6	18 700	31 100	17 000	LRT 253017
	NAFW 254026	135	25	40	26	0.3	30	34.5	38	27	29	30.6	24 200	43 400	17 000	LRT 253026
	NAF 254216	94	25	42	16	0.3	30	35.4	40	27	29	30.6	20 800	28 400	17 000	LRT 253016
	NAFW 254232	186	25	42	32	0.3	30	35.4	40	27	29	30.6	35 700	56 800	17 000	LRT 253032

Notes(1) Minimum allowable value of chamfer dimension *r*
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 50% of this value is allowable.
Remarks1. RNAF has no oil hole. RNAFW is provided with an oil groove and an oil hole on the outer ring.
2. No grease is prepacked. Perform proper lubrication.

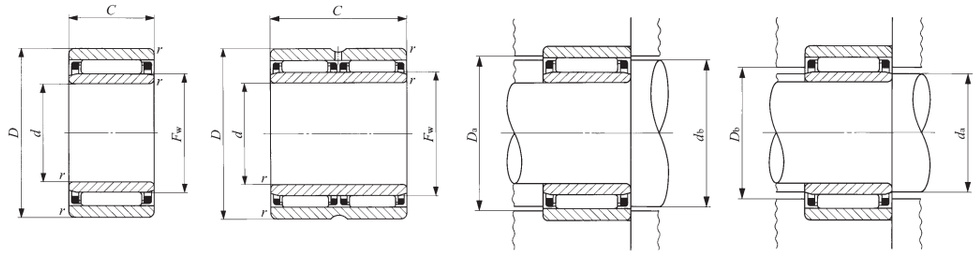
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1mm=0.03937inch

NEEDLE ROLLER BEARINGS WITH SEPARABLE CAGE

With Inner Ring







NAF NAFW

Shaft dia. 30 — 65mm

Shaft dia. mm	Identification number	Mass (Ref.) g	Boundary dimensions mm d	D	C	$r_{s\,min}$ (1)	F_w	Standard mounting dimensions mm d_b	D_a Max.	d_a Min.	d_a Max.	D_b	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed (2) rpm	Assembled inner ring
30	NAF 304517	101	30	45	17	0.3	35	39.5	43	32	34	35.6	20 500	36 900	14 000	LRT 303517
	NAFW 304526	155	30	45	26	0.3	35	39.5	43	32	34	35.6	26 600	51 500	14 000	LRT 303526
	NAF 304716	107	30	47	16	0.3	35	40.4	45	32	34	35.6	23 100	33 900	14 000	LRT 303516
	NAFW 304732	215	30	47	32	0.3	35	40.4	45	32	34	35.6	39 500	67 800	14 000	LRT 303532
35	NAF 355017	115	35	50	17	0.3	40	43.5	48	37	39	40.8	22 200	42 700	12 000	LRT 354017
	NAFW 355034	230	35	50	34	0.3	40	43.5	48	37	39	40.8	38 000	85 400	12 000	LRT 354034
	NAF 355520	186	35	55	20	0.3	40	45.2	53	37	39	40.8	31 400	48 000	12 000	LRT 354020
	NAFW 355540	375	35	55	40	0.3	40	45.2	53	37	39	40.8	53 900	96 000	12 000	LRT 354040
40	NAF 405517	128	40	55	17	0.3	45	48.5	53	42	44	45.8	23 300	47 100	11 000	LRT 404517
	NAFW 405534	255	40	55	34	0.3	45	48.5	53	42	44	45.8	39 900	94 200	11 000	LRT 404534
	NAF 406220	235	40	62	20	0.3	45	50.9	60	42	44	45.8	33 200	53 300	11 000	LRT 404520
	NAFW 406240	475	40	62	40	0.3	45	50.9	60	42	44	45.8	56 900	107 000	11 000	LRT 404540
45	NAF 456220	196	45	62	20	0.3	50	54.2	60	47	49	50.8	27 100	59 300	10 000	LRT 455020
	NAFW 456240	390	45	62	40	0.3	50	54.2	60	47	49	50.8	46 400	119 000	10 000	LRT 455040
	NAF 457220	340	45	72	20	1	55	60.9	67	50	54	55.8	37 400	66 400	9 000	LRT 455520
	NAFW 457240	685	45	72	40	1	55	60.9	67	50	54	55.8	64 100	133 000	9 000	LRT 455540
50	NAF 506820	230	50	68	20	0.3	55	59.5	66	52	54	55.8	28 600	66 000	9 000	LRT 505520
	NAFW 506840	465	50	68	40	0.3	55	59.5	66	52	54	55.8	49 000	132 000	9 000	LRT 505540
	NAF 507820	390	50	78	20	1	60	66.3	73	55	59	60.8	38 900	71 700	8 500	LRT 506020
	NAFW 507840	775	50	78	40	1	60	66.3	73	55	59	60.8	66 700	143 000	8 500	LRT 506040
55	NAF 558530	690	55	85	30	1.5	65	72	77	63	63.5	66	59 300	127 000	7 500	LRT 556530
	NAFW 558560	1 380	55	85	60	1.5	65	72	77	63	63.5	66	102 000	255 000	7 500	LRT 556560
60	NAF 609030	740	60	90	30	1.5	70	77	82	68	68.5	71	61 200	136 000	7 000	LRT 607030
	NAFW 609060	1 480	60	90	60	1.5	70	77	82	68	68.5	71	105 000	272 000	7 000	LRT 607060
65	NAF 659530	790	65	95	30	1.5	75	82	87	73	73.5	76	63 100	144 000	6 500	LRT 657530
	NAFW 659560	1 580	65	95	60	1.5	75	82	87	73	73.5	76	108 000	289 000	6 500	LRT 657560

Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 50% of this value is allowable.
Remarks1. RNAF has no oil hole. RNAFW is provided with an oil groove and an oil hole on the outer ring.
2. No grease is prepacked. Perform proper lubrication.

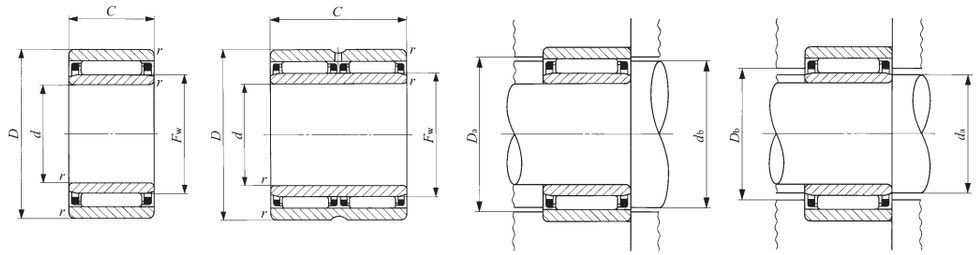
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1mm=0.03937inch

NEEDLE ROLLER BEARINGS WITH SEPARABLE CAGE

With Inner Ring







NAF NAFW

Shaft dia. 70 — 90mm

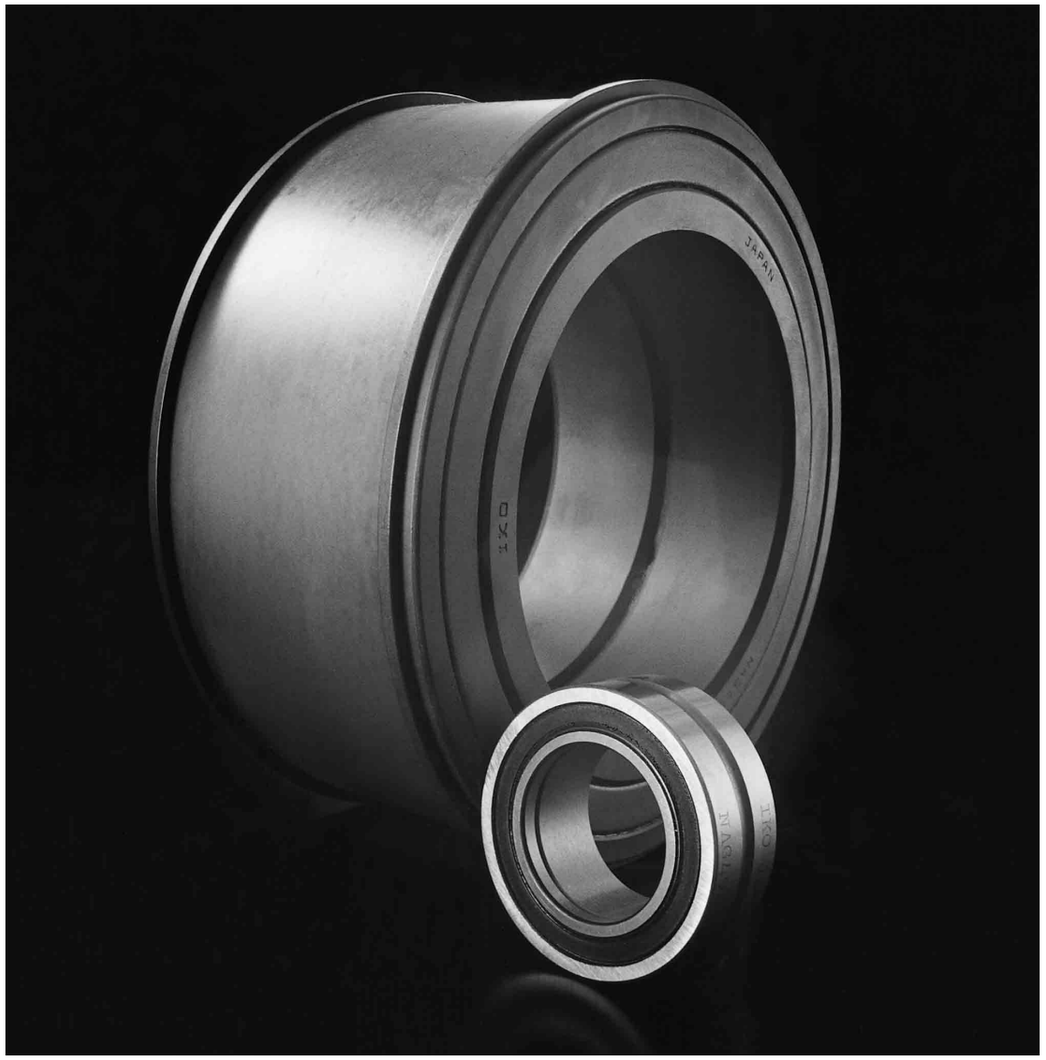
Shaft dia. mm	Identification number	Mass (Ref.) g	Boundary dimensions mm d	D	C	$r_{s\,min}$ (1)	F_w	Standard mounting dimensions mm d_b	D_a Max.	d_a Min.	d_a Max.	D_b	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed (2) rpm	Assembled inner ring
70	**NAF 7010030**	835	70	100	30	1.5	80	87	92	78	78.5	81	65 000	153 000	6 000	**LRT 708030-1**
	NAFW 7010060	1 680	70	100	60	1.5	80	87	92	78	78.5	81	111 000	306 000	6 000	**LRT 708060**
75	**NAF 7510530**	885	75	105	30	1.5	85	92	97	83	83.5	86	66 600	161 000	6 000	**LRT 758530-1**
80	**NAF 8011030**	935	80	110	30	1.5	90	97	102	88	88.5	91	69 600	174 000	5 500	**LRT 809030-1**
85	**NAF 8511530**	985	85	115	30	1.5	95	102	107	93	93.5	96	70 900	182 000	5 500	**LRT 859530**
90	**NAF 9012030**	1 040	90	120	30	1.5	100	107	112	98	98.5	101	72 500	191 000	4 500	**LRT 9010030**

Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 50% of this value is allowable.
Remarks1. RNAF has no oil hole. RNAFW is provided with an oil groove and an oil hole on the outer ring.
2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

ROLLER BEARINGS

- **Caged Roller Bearings**
- **Full Complement Roller Bearings**
- **Roller Bearings for Sheaves**

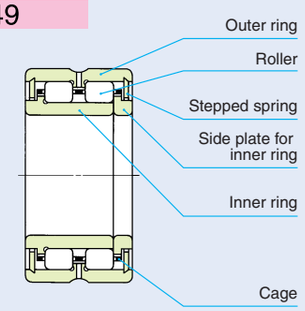


Structure and Features

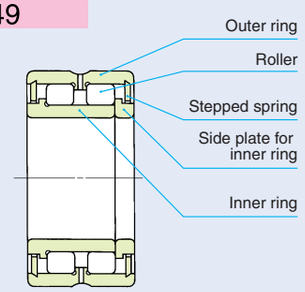
IKO Roller Bearings in which rollers are incorporated in two rows are non-separable heavy-duty bearings. They can withstand not only radial loads but axial loads as well, which are supported at the contacts between the shoulders of inner and outer rings and the end faces of rollers. Therefore, they are most suitable for use at the fixing side of a shaft. Like needle roller bearings, they are also compact.
Roller bearings include the caged type, full complement type and the type for sheaves, and any bearings suitable for the operating conditions can be selected. In particular, these bearings are used for heavy-duty machines such as construction machinery, and industrial machinery.

Structures of Roller Bearings

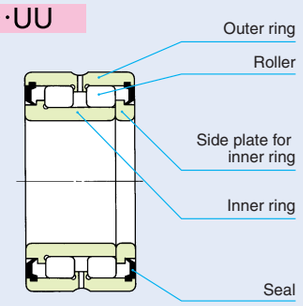
NAU49



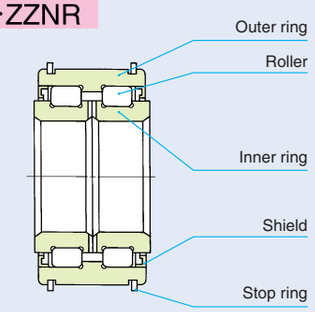
NAG49



NAG49···UU



NAS50···ZZNR



Types

The types of Roller Bearings shown in Table 1 are available.

Table 1 Type of bearing

Type / Series	Caged type	Full complement type	For sheaves
Standard	NAU49 TRU	NAG49	——
With seal	NAU49…UU TRU…UU	NAG49…UU	NAS50…UUNR
With shield	——	——	NAS50…ZZNR

Caged Roller Bearings

These bearings are suitable for high-speed rotations and fluctuating loads. Also, as the axial distance between the double-row rollers is comparatively large, large moment loads can be supported.
Caged roller bearings with seal incorporate seals on both sides. Synthetic resin rubber seals are excellent in the prevention of dust penetration and grease leakage, providing an excellent sealing effect.

Full Complement Roller Bearings

These bearings are suitable for low-speed rotations or oscillating motions and heavy loads. Similar to the caged type, the structure is advantageous for supporting moment loads.
The bearings with seal incorporate seals on both sides.

Roller Bearings for Sheaves

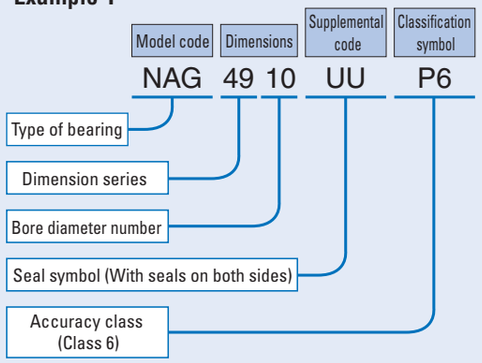
These bearings are the double-row full complement type with a low sectional height designed for use in sheaves. There are two types; the sealed type and the shield type. They can withstand heavy radial loads and shock loads at comparatively low-speed rotations, and can also withstand axial loads.
They can easily be fixed axially to sheaves using the stop rings of the outer ring. As the width of the inner ring is designed to be larger than that of the outer ring, they require no spacer between sheaves. The structure is stable because the double-row rollers can withstand the moment loads caused by rope transition.
The surfaces of these bearings are treated to have high corrosion resistance.

Identification Number

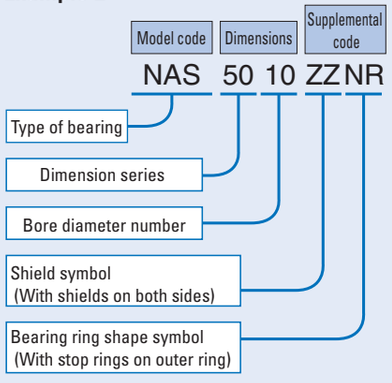
The identification number of Roller Bearings consists of a model code, dimensions, any supplemental codes and a classification symbol. The arrangement examples are shown below.

Examples of identification number

Example 1



Example 2



Accuracy

Roller Bearings are manufactured in accordance with JIS (See page A31.). A side plate for inner ring is assembled on one side of caged or full complement roller bearings. The tolerance of bore diameter of the side plate is shown below. Tolerances of Roller Bearings for Sheaves represent the values before surface treatment. The tolerance of internal distance between cir-clips is shown below.
Tolerance of bore diameter of the side plate d: E7
Tolerance of internal distance between cir-clips C_1: 0～+0.4mm

Clearance

Roller Bearings are manufactured to the CN clearance shown in Table 18 on page A37. However, Roller Bearings for Sheaves are manufactured so that proper operating clearances are obtained after being mounted with a specified fit.

Fit

The recommended fits for Roller Bearings are shown in Tables 21 to 22 on pages A41 and A42. The recommended fits for Roller Bearings for Sheaves are shown in Table 2.

Lubrication

Bearings with prepacked grease are shown in Table 3. For Caged Roller Bearings and Full Complement Roller Bearings, ALVANIA GREASE S2 (SHELL) is prepacked as the lubricating grease. For Roller Bearings for Sheaves, ALVANIA GREASE EP2 (SHELL) is prepacked as the lubricating grease.
In the case of bearings without prepacked grease, perform proper lubrication for use. Operating without lubrication will increase the wear of the rolling contact surfaces and shorten their lirees.

Oil Hole

The number of oil holes of the inner and outer rings is shown in Table 4.

Operating Temperature Range

The operating temperature range for Roller Bearings is −20℃～+120℃. However, the maximum allowable temperature for Roller Bearings for Sheaves is +110℃.

Table 2 Recommended fits for Roller Bearings for Sheaves

Tolerance class of shaft	Tolerance class of housing bore
g6	N7

Table 3 Bearings with prepacked grease ○: With prepacked grease ×: Without prepacked grease

Type		Standard	With seals	With shields
Caged type	NAU, TRU	×	○	−
Full complement type	NAG	×	○	−
For sheaves	NAS	−	○	○

Table 4 Number of oil holes of the inner ring and outer ring

Type			Number of oil holes of the outer ring			Number of oil holes of the inner ring
		Nominal bore diameter d mm	Standard	With seals	With shields	
Caged type	NAU	$d \leqq 17$	0	0	—	0
		$17 < d$	2	2		
	TRU		2	2	—	0
Full complement type	NAG	$d \leqq 17$	0	0	—	0
		$17 < d$	2	2		
For sheaves	NAS		—	0	0	2

Remark The bearings with oil holes are also provided with an oil groove.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

Axial Load Capacity

Axial load capacity is not determined from the basic dynamic load rating based on rolling fatigue, but is determined by the amount of heat generated by sliding contact between the ends of rollers and guide shoulders of the inner and outer rings. It is therefore limited by the load conditions, sliding speeds, lubrication methods, etc.

The axial load capacity of Roller Bearings is obtained from the following equation.

If the axial load increases in comparison with the radial load, it will start to interfere with the smooth rolling motion. The axial load should therefore be within 20% of the radial load.

$$C_A = f_v \, a \, f_A \quad \cdots\cdots (1)$$

where, C_A : Axial load capacity N

f_v : Speed correction factor

f_v is obtained from Fig.2 by calculating the $d_m n$ value.

$d_m n = d_m \times n$

d_m : Mean value of bearing bore and outside diameters mm

$$\left(d_m \fallingdotseq \frac{d+D}{2} \right)$$

n : Rotational speed rpm

When $d_m n \leqq 1000$, $f_v = 1$.

a : Value determined by type of bearing (See Table 5.)

f_A : Axial load capacity factor (See Fig.1.)

Table 5 Value by type of bearing

Type of bearing	a
NAS 50	1
NAG 49	0.78
NAU 49, TRU	0.7

Calculation example

When a roller bearing for sheaves NAS 5016 ZZ NR is run at n = 250 rpm under grease lubrication and subjected to an intermittent axial load, the axial load capacity is calculated as follows.

As the bearing bore diameter is 80 mm, f_A = 18000 is obtained from the axial load capacity line of Fig. 1 (ii).

$a = 1$

$$d_m \fallingdotseq \frac{80+125}{2} = 102.5$$

$d_m n = 102.5 \text{ x } 250 \fallingdotseq 25600$

From Fig. 2, $f_v \fallingdotseq 0.87$

Therefore, the axial load capacity C_A is obtained.

$C_A = f_v \, a \, f_A = 0.87 \times 1 \times 18000 \fallingdotseq 15700$ N

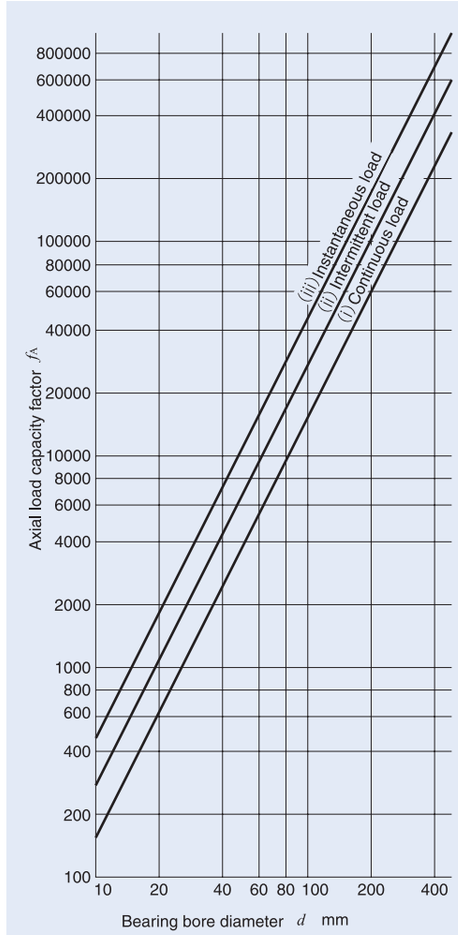


Fig. 1 Axial load capacity factor

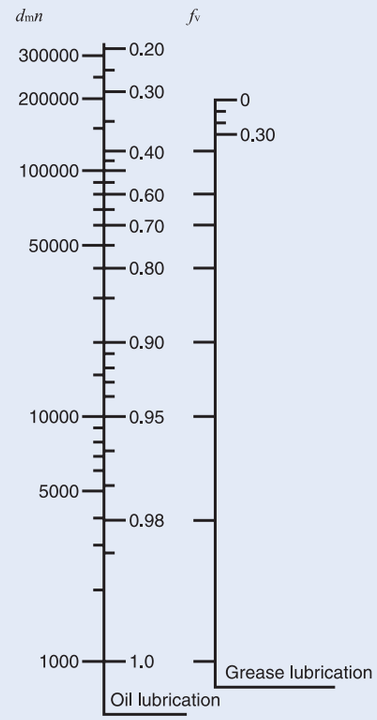


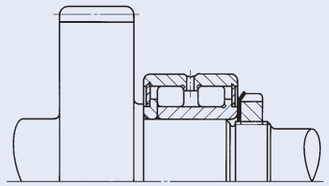
Fig. 2 Speed correction factor

Mounting

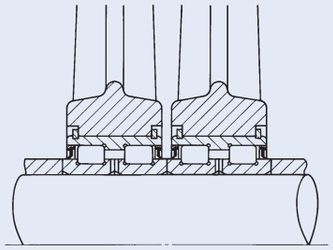
Unlike needle roller bearings, Caged and Full Complement Roller Bearings are non-separable.

As shown in Fig. 3 (1), the inner ring should be press-fitted until it makes close contact with the shaft shoulder, and fixed axially with a nut. Dimensions of the shoulders of the shaft and housing should be based on J and E_W shown in the table of dimensions, respectively.

In the case of Roller Bearings for Sheaves, as shown in Fig. 3 (2), the outer ring should be fixed by stop rings after being press-fitted into the sheaves, and the inner ring should be fixed securely in the axial direction.



(1)



(2)

Fig. 3 Mounting examples

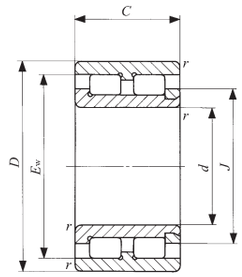
1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

ROLLER BEARINGS

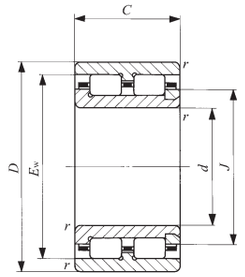
Caged Roller Bearings
Full Complement Roller Bearings



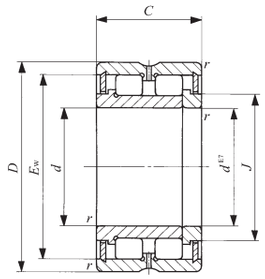




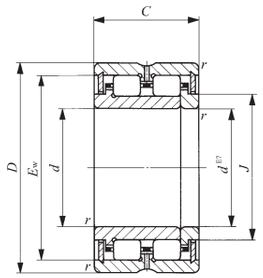
NAG49
($d \leqq 17$)



NAU49
($d \leqq 17$)



NAG49



NAU49 TRU

Shaft dia. 10 — 35mm

Shaft dia. mm	Identification number: Full complement type	Identification number: Caged type		Mass (Ref.) g	Boundary dimensions mm: d	D	C	$r_{s\,min}$ (1)	J	E_w	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed (2) rpm
10	**NAG 4900**	—	—	25.5	10	22	13	0.3	15.5	18.5	9 650	10 800	17 000
	—	**NAU 4900**	—	24.5	10	22	13	0.3	15.5	18.5	6 580	6 470	30 000
12	**NAG 4901**	—	—	28.5	12	24	13	0.3	17	20	10 300	12 000	15 000
	—	**NAU 4901**	—	27.5	12	24	13	0.3	17	20	6 950	7 120	25 000
15	**NAG 4902**	—	—	38	15	28	13	0.3	21	24	11 800	15 200	12 000
	—	**NAU 4902**	—	36.5	15	28	13	0.3	21	24	7 950	9 020	20 000
	—	—	**TRU 153320**	80.5	15	33	20	0.3	19.5	27	10 400	10 400	20 000
17	**NAG 4903**	—	—	41	17	30	13	0.3	22.5	25.5	12 300	16 500	11 000
	—	**NAU 4903**	—	39.5	17	30	13	0.3	22.5	25.5	8 240	9 670	19 000
	—	—	**TRU 173425**	100	17	34	25	0.3	21.5	29.5	18 000	21 600	18 000
20	**NAG 4904**	—	—	76.5	20	37	17	0.3	24	31.5	15 600	18 900	9 500
	—	**NAU 4904**	—	76	20	37	17	0.3	24	31.5	10 700	11 300	16 000
	—	—	**TRU 203820**	96.5	20	38	20	0.3	25	32.5	12 100	13 400	16 000
	—	—	**TRU 203825**	122	20	38	25	0.3	25	32.5	18 700	23 600	16 000
25	**NAG 4905**	—	—	89.5	25	42	17	0.3	29.5	37	17 500	23 200	7 500
	—	**NAU 4905**	—	89	25	42	17	0.3	29.5	37	11 900	13 900	13 000
	—	—	**TRU 254425**	154	25	44	25	0.3	30.5	38	21 000	28 900	13 000
28	—	—	**TRU 284530**	173	28	45	30	0.3	31.5	39.5	28 700	43 800	12 000
30	**NAG 4906**	—	—	103	30	47	17	0.3	34	41.5	19 400	27 600	6 500
	—	**NAU 4906**	—	102	30	47	17	0.3	34	41.5	13 000	16 200	12 000
	—	—	**TRU 304830**	197	30	48	30	0.3	35	42.5	29 400	46 600	11 000
32	—	—	**TRU 325230**	260	32	52	30	0.6	38	46	29 800	44 200	10 000
35	**NAG 4907**	—	—	172	35	55	20	0.6	40	49	28 700	43 800	5 500
	—	**NAU 4907**	—	168	35	55	20	0.6	40	49	19 500	26 300	10 000
	—	—	**TRU 355630**	270	35	56	30	0.6	40	49	32 200	49 800	10 000

Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable. Considering that the axial load acts under practical operating conditions, up to 1/10 of this value is recommended for actual use.

Remarks1. The NAG and NAU series with a bore diameter d of 17 mm or less have no oil hole. In others, the outer ring has an oil groove and two oil holes.
2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

ROLLER BEARINGS

Caged Roller Bearings
Full Complement Roller Bearings



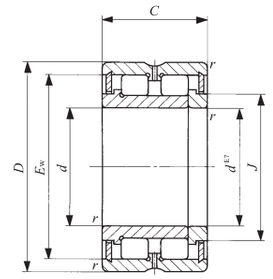


Shaft dia. 40 — 80mm

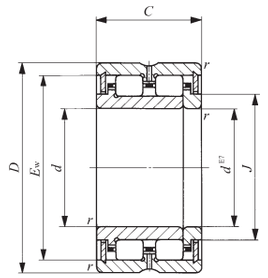
Shaft dia. mm	Identification number: Full complement type	Identification number: Caged type		Mass (Ref.) g	Boundary dimensions mm: d	D	C	$r_{s\,min}$ (1)	J	E_w
40	**NAG 4908**	—	—	225	40	62	22	0.6	46	56
	—	—	**TRU 405930**	265	40	59	30	0.6	45	52.5
	—	**NAU 4908**	—	220	40	62	22	0.6	46	56
42	—	—	**TRU 426230**	290	42	62	30	0.6	48	56.5
45	**NAG 4909**	—	—	265	45	68	22	0.6	51	61
	—	—	**TRU 456430**	295	45	64	30	0.6	50.5	58.5
	—	**NAU 4909**	—	260	45	68	22	0.6	51	61
50	**NAG 4910**	—	—	270	50	72	22	0.6	55.5	65.5
	—	**NAU 4910**	—	265	50	72	22	0.6	55.5	65.5
	—	—	**TRU 507745**	710	50	77	45	1	58	69
55	**NAG 4911**	—	—	395	55	80	25	1	61.5	72.5
	—	**NAU 4911**	—	385	55	80	25	1	61.5	72.5
	—	—	**TRU 558138**	615	55	81	38	1	61.5	72.5
60	**NAG 4912**	—	—	425	60	85	25	1	67	77.5
	—	**NAU 4912**	—	415	60	85	25	1	67	77.5
	—	—	**TRU 608945**	880	60	89	45	1	69.5	81.5
65	**NAG 4913**	—	—	455	65	90	25	1	72	83
	—	**NAU 4913**	—	440	65	90	25	1	72	83
70	**NAG 4914**	—	—	725	70	100	30	1	79	91.5
	—	**NAU 4914**	—	705	70	100	30	1	79	91.5
75	**NAG 4915**	—	—	775	75	105	30	1	83.5	95.5
	—	**NAU 4915**	—	750	75	105	30	1	83.5	95.5
	—	—	**TRU 7510845**	1 240	75	108	45	1	85.5	98.5
80	**NAG 4916**	—	—	815	80	110	30	1	89.5	102
	—	**NAU 4916**	—	790	80	110	30	1	89.5	102

Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable. Considering that the axial load acts under practical operating conditions, up to 1/10 of this value is recommended for actual use.

Remarks1. The outer ring has an oil groove and two oil holes.
2. No grease is prepacked. Perform proper lubrication.



NAG49



NAU49 TRU

Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
34 600	49 500	5 000
34 700	62 500	8 500
23 400	29 400	8 500
34 600	57 800	8 000
36 400	54 700	4 500
32 600	59 700	8 000
24 800	32 800	8 000
38 200	59 900	4 000
26 200	36 200	7 000
75 700	134 000	7 000
48 100	77 700	3 500
33 000	47 000	6 500
61 400	104 000	6 500
50 300	84 300	3 500
34 700	51 400	6 000
88 100	152 000	6 000
53 200	93 000	3 000
36 900	57 100	5 500
77 700	139 000	3 000
53 700	84 600	5 000
80 000	146 000	2 500
54 800	88 200	5 000
103 000	190 000	4 500
83 000	157 000	2 500
57 200	95 500	4 500

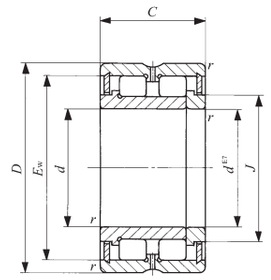
1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

ROLLER BEARINGS

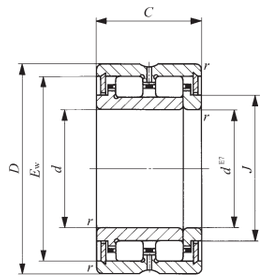
Caged Roller Bearings
Full Complement Roller Bearings







NAG49



NAU49 TRU

Shaft dia. 85 — 140mm

Shaft dia. mm	Identification number: Full complement type	Identification number: Caged type		Mass (Ref.) g	Boundary dimensions mm: d	D	C	$r_{s\,min}$ (1)	J	E_w	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed (2) rpm
85	NAG 4917	—	—	1 190	85	120	35	1.5	96	110	111 000	200 000	2 500
	—	—	TRU 8511850	1 530	85	118	50	1	94.5	107.5	114 000	222 000	4 000
	—	NAU 4917	—	1 150	85	120	35	1.5	96	110	75 400	120 000	4 000
	—	—	TRU 8512045	1 500	85	120	45	1.5	96.5	110	110 000	215 000	4 000
90	NAG 4918	—	—	1 250	90	125	35	1.5	101	115.5	114 000	211 000	2 500
	—	NAU 4918	—	1 210	90	125	35	1.5	101	115.5	79 500	130 000	4 000
	—	—	TRU 9012550	1 740	90	125	50	1.5	101	114	119 000	240 000	4 000
95	NAG 4919	—	—	1 300	95	130	35	1.5	106	120.5	117 000	222 000	2 000
	—	NAU 4919	—	1 270	95	130	35	1.5	106	120.5	81 000	136 000	4 000
100	NAG 4920	—	—	1 850	100	140	40	1.5	114.5	129.5	152 000	292 000	2 000
	—	—	TRU 10013550	1 900	100	135	50	1.5	112	125.5	124 000	264 000	3 500
	—	NAU 4920	—	1 770	100	140	40	1.5	114.5	129.5	106 000	181 000	3 500
105	—	—	TRU 10515350	2 890	105	153	50	1.5	120	138	159 000	286 000	3 500
110	NAG 4922	—	—	2 010	110	150	40	1.5	123	138.5	161 000	322 000	1 900
	—	NAU 4922	—	1 930	110	150	40	1.5	123	138.5	113 000	200 000	3 500
120	NAG 4924	—	—	2 780	120	165	45	1.5	136	153.5	208 000	431 000	1 700
	—	NAU 4924	—	2 680	120	165	45	1.5	136	153.5	146 000	268 000	3 000
125	—	—	TRU 12517860	4 490	125	178	60	1.5	143.5	162	211 000	408 000	3 000
130	NAG 4926	—	—	3 750	130	180	50	2	147	165.5	240 000	495 000	1 600
	—	NAU 4926	—	3 610	130	180	50	2	147	165.5	166 000	304 000	2 500
135	—	—	TRU 13518860	4 790	135	188	60	1.5	154	172.5	220 000	442 000	2 500
140	NAG 4928	—	—	3 990	140	190	50	2	157.5	176	249 000	531 000	1 500
	—	NAU 4928	—	3 840	140	190	50	2	157.5	176	174 000	327 000	2 500

Notes(1) Minimum allowable value of chamfer dimension *r*
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable. Considering that the axial load acts under practical operating conditions, up to 1/10 of this value is recommended for actual use.

Remarks1. The outer ring has an oil groove and two oil holes.
2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

ROLLER BEARINGS

Caged Roller Bearings With Seal
Full Complement Roller Bearings With Seal



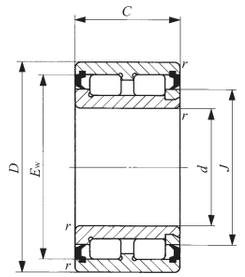


Shaft dia. 10 — 40mm

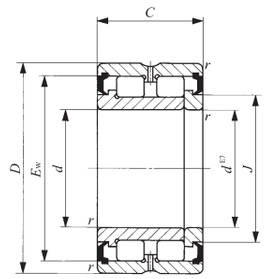
Shaft dia. mm	Identification number: Full complement type	Identification number: Caged type		Mass (Ref.) g	Boundary dimensions mm: d	D	C	$r_{s\,min}$ (1)	J
10	NAG 4900UU	—	—	25.5	10	22	13	0.3	15.5
12	NAG 4901UU	—	—	28.5	12	24	13	0.3	17
15	NAG 4902UU	—	—	38	15	28	13	0.3	21
	—	—	TRU 153320UU	80.5	15	33	20	0.3	19.5
17	NAG 4903UU	—	—	41	17	30	13	0.3	22.5
	—	—	TRU 173425UU	100	17	34	25	0.3	21.5
20	NAG 4904UU	—	—	76.5	20	37	17	0.3	24
	—	NAU 4904UU	—	76	20	37	17	0.3	24
	—	—	TRU 203820UU	96.5	20	38	20	0.3	25
	—	—	TRU 203825UU	122	20	38	25	0.3	25
25	NAG 4905UU	—	—	89.5	25	42	17	0.3	29.5
	—	NAU 4905UU	—	89	25	42	17	0.3	29.5
	—	—	TRU 254425UU	154	25	44	25	0.3	30.5
28	—	—	TRU 284530UU	173	28	45	30	0.3	31.5
30	NAG 4906UU	—	—	103	30	47	17	0.3	34
	—	NAU 4906UU	—	102	30	47	17	0.3	34
	—	—	TRU 304830UU	197	30	48	30	0.3	35
32	—	—	TRU 325230UU	260	32	52	30	0.6	38
35	NAG 4907UU	—	—	172	35	55	20	0.6	40
	—	NAU 4907UU	—	168	35	55	20	0.6	40
	—	—	TRU 355630UU	270	35	56	30	0.6	40
40	NAG 4908UU	—	—	225	40	62	22	0.6	46
	—	—	TRU 405930UU	265	40	59	30	0.6	45
	—	NAU 4908UU	—	220	40	62	22	0.6	46

Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to grease lubrication. Considering that the axial load acts under practical operating conditions, up to 1/10 of this value is recommended for actual use.

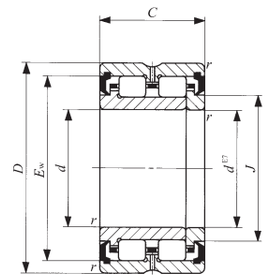
Remarks1. The NAG and NAU series with a bore diameter, d, of 17 mm or less have no oil hole. In others, the outer ring has an oil groove and two oil holes.
2. The bearings with seals are provided with prepacked grease.



NAG49···UU
($d \leqq 17$)



NAG49···UU



NAU49···UU
TRU···UU

E_w	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
19.5	9 650	10 800	10 000
21	10 300	12 000	9 000
25	11 800	15 200	7 000
27	10 400	10 400	9 500
26.5	12 300	16 500	6 500
29.5	18 000	21 600	8 500
31.5	15 600	18 900	5 500
31.5	10 700	11 300	8 000
32.5	12 100	13 400	7 500
32.5	18 700	23 600	7 500
37	17 500	23 200	4 500
37	11 900	13 900	6 500
38	21 000	28 900	6 000
39.5	28 700	43 800	6 000
41.5	19 400	27 600	4 000
41.5	13 000	16 200	5 500
42.5	29 400	46 600	5 500
46	29 800	44 200	5 000
49	28 700	43 800	3 500
49	19 500	26 300	4 500
49	32 200	49 800	4 500
56	34 600	49 500	3 000
52.5	34 700	62 500	4 000
56	23 400	29 400	4 000

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

ROLLER BEARINGS

Caged Roller Bearings With Seal
Full Complement Roller Bearings With Seal

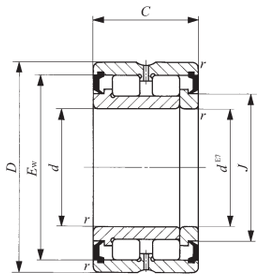




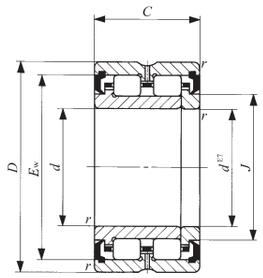
Shaft dia. 42 — 80mm

Shaft dia. mm	Identification number: Full complement type	Identification number: Caged type		Mass (Ref.) g	Boundary dimensions mm: d	D	C	$r_{s\,min}$ (1)	J
42	—	—	TRU 426230UU	290	42	62	30	0.6	48
45	NAG 4909UU	—	—	265	45	68	22	0.6	51
	—	—	TRU 456430UU	295	45	64	30	0.6	50.5
	—	NAU 4909UU	—	260	45	68	22	0.6	51
50	NAG 4910UU	—	—	270	50	72	22	0.6	55.5
	—	NAU 4910UU	—	265	50	72	22	0.6	55.5
	—	—	TRU 507745UU	710	50	77	45	1	58
55	NAG 4911UU	—	—	395	55	80	25	1	61.5
	—	NAU 4911UU	—	385	55	80	25	1	61.5
	—	—	TRU 558138UU	615	55	81	38	1	61.5
60	NAG 4912UU	—	—	425	60	85	25	1	67
	—	NAU 4912UU	—	415	60	85	25	1	67
	—	—	TRU 608945UU	880	60	89	45	1	69.5
65	NAG 4913UU	—	—	455	65	90	25	1	72
	—	NAU 4913UU	—	440	65	90	25	1	72
70	NAG 4914UU	—	—	725	70	100	30	1	79
	—	NAU 4914UU	—	705	70	100	30	1	79
75	NAG 4915UU	—	—	775	75	105	30	1	83.5
	—	NAU 4915UU	—	750	75	105	30	1	83.5
	—	—	TRU 7510845UU	1 240	75	108	45	1	85.5
80	NAG 4916UU	—	—	815	80	110	30	1	89.5
	—	NAU 4916UU	—	790	80	110	30	1	89.5

Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to grease lubrication. Considering that the axial load acts under practical operating conditions, up to 1/10 of this value is recommended for actual use.
Remarks1. The outer ring has an oil groove and two oil holes.
2. The bearings with seals are provided with prepacked grease.



NAG49···UU



NAU49···UU
TRU···UU

E

NAG
NAU
TRU
NAS

E_w	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
56.5	34 600	57 800	4 000
61	36 400	54 700	2 500
58.5	32 600	59 700	3 500
61	24 800	32 800	3 500
65.5	38 200	59 900	2 500
65.5	26 200	36 200	3 500
69	75 700	134 000	3 500
72.5	48 100	77 700	2 000
72.5	33 000	47 000	3 000
72.5	61 400	104 000	3 000
77.5	50 300	84 300	2 000
77.5	34 700	51 400	3 000
81.5	88 100	152 000	3 000
83	53 200	93 000	1 900
83	36 900	57 100	2 500
91.5	77 700	139 000	1 800
91.5	53 700	84 600	2 500
95.5	80 000	146 000	1 700
95.5	54 800	88 200	2 500
98.5	103 000	190 000	2 000
102	83 000	157 000	1 600
102	57 200	95 500	2 000

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

ROLLER BEARINGS

Caged Roller Bearings With Seal
Full Complement Roller Bearings With Seal

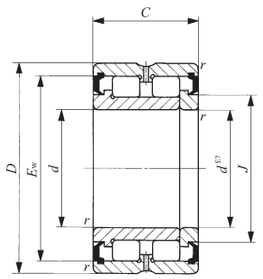




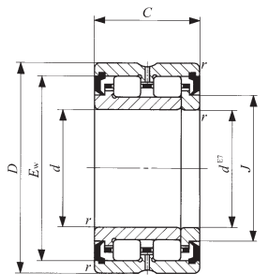
Shaft dia. 85 — 140mm

Shaft dia. mm	Identification number: Full complement type	Identification number: Caged type		Mass (Ref.) g	Boundary dimensions mm: d	D	C	$r_{s\,min}$ (1)	J
85	**NAG 4917UU**	—	—	1 190	85	120	35	1.5	96
	—	—	**TRU 8511850UU**	1 530	85	118	50	1	94.5
	—	**NAU 4917UU**	—	1 150	85	120	35	1.5	96
	—	—	**TRU 8512045UU**	1 500	85	120	45	1.5	96.5
90	**NAG 4918UU**	—	—	1 250	90	125	35	1.5	101
	—	**NAU 4918UU**	—	1 210	90	125	35	1.5	101
	—	—	**TRU 9012550UU**	1 740	90	125	50	1.5	101
95	**NAG 4919UU**	—	—	1 300	95	130	35	1.5	106
	—	**NAU 4919UU**	—	1 270	95	130	35	1.5	106
100	**NAG 4920UU**	—	—	1 850	100	140	40	1.5	114.5
	—	—	**TRU 10013550UU**	1 900	100	135	50	1.5	112
	—	**NAU 4920UU**	—	1 770	100	140	40	1.5	114.5
105	—	—	**TRU 10515350UU**	2 890	105	153	50	1.5	120
110	**NAG 4922UU**	—	—	2 010	110	150	40	1.5	123
	—	**NAU 4922UU**	—	1 930	110	150	40	1.5	123
120	**NAG 4924UU**	—	—	2 780	120	165	45	1.5	136
	—	**NAU 4924UU**	—	2 680	120	165	45	1.5	136
125	—	—	**TRU 12517860UU**	4 490	125	178	60	1.5	143.5
130	**NAG 4926UU**	—	—	3 750	130	180	50	2	147
	—	**NAU 4926UU**	—	3 610	130	180	50	2	147
135	—	—	**TRU 13518860UU**	4 790	135	188	60	1.5	154
140	**NAG 4928UU**	—	—	3 990	140	190	50	2	157.5
	—	**NAU 4928UU**	—	3 840	140	190	50	2	157.5

Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to grease lubrication. Considering that the axial load acts under practical operating conditions, up to 1/10 of this value is recommended for actual use.
Remarks1. The outer ring has an oil groove and two oil holes.
2. The bearings with seals are provided with prepacked grease.



NAG49···UU



NAU49···UU
TRU···UU

E
NAG
NAU
TRU
NAS

E_w	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
110	111 000	200 000	1 500
107.5	114 000	222 000	2 000
110	75 400	120 000	2 000
110	110 000	215 000	2 000
115.5	114 000	211 000	1 400
115.5	79 500	130 000	1 900
114	119 000	240 000	1 900
120.5	117 000	222 000	1 300
120.5	81 000	136 000	1 800
129.5	152 000	292 000	1 200
125.5	124 000	264 000	1 700
129.5	106 000	181 000	1 700
138	159 000	286 000	1 600
138.5	161 000	322 000	1 100
138.5	113 000	200 000	1 600
153.5	208 000	431 000	1 000
153.5	146 000	268 000	1 400
162	211 000	408 000	1 400
165.5	240 000	495 000	950
165.5	166 000	304 000	1 300
172.5	220 000	442 000	1 300
176	249 000	531 000	900
176	174 000	327 000	1 200

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

ROLLER BEARINGS

Roller Bearings for Sheaves





Shaft dia. 40 — 170mm

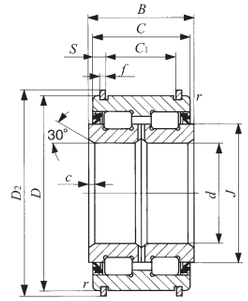
Shaft dia. mm	Identification number		Mass (Ref.)	Boundary dimensions mm						
	Sealed type	Shield type	kg	d	D	D_2	B	C	C_1	S
40	**NAS 5008UUNR**	**NAS 5008ZZNR**	0.55	40	68	71.8	38	37	28	4.5
45	**NAS 5009UUNR**	**NAS 5009ZZNR**	0.70	45	75	78.8	40	39	30	4.5
50	**NAS 5010UUNR**	**NAS 5010ZZNR**	0.75	50	80	83.8	40	39	30	4.5
55	**NAS 5011UUNR**	**NAS 5011ZZNR**	1.15	55	90	94.8	46	45	34	5.5
60	**NAS 5012UUNR**	**NAS 5012ZZNR**	1.20	60	95	99.8	46	45	34	5.5
65	**NAS 5013UUNR**	**NAS 5013ZZNR**	1.30	65	100	104.8	46	45	34	5.5
70	**NAS 5014UUNR**	**NAS 5014ZZNR**	1.90	70	110	114.5	54	53	42	5.5
75	**NAS 5015UUNR**	**NAS 5015ZZNR**	2.00	75	115	119.5	54	53	42	5.5
80	**NAS 5016UUNR**	**NAS 5016ZZNR**	2.65	80	125	129.5	60	59	48	5.5
85	**NAS 5017UUNR**	**NAS 5017ZZNR**	2.80	85	130	134.5	60	59	48	5.5
90	**NAS 5018UUNR**	**NAS 5018ZZNR**	3.70	90	140	145.4	67	66	54	6
95	**NAS 5019UUNR**	**NAS 5019ZZNR**	3.90	95	145	150.4	67	66	54	6
100	**NAS 5020UUNR**	**NAS 5020ZZNR**	4.05	100	150	155.4	67	66	54	6
110	**NAS 5022UUNR**	**NAS 5022ZZNR**	6.50	110	170	175.4	80	79	65	7
120	**NAS 5024UUNR**	**NAS 5024ZZNR**	6.95	120	180	188.4	80	79	65	7
130	**NAS 5026UUNR**	**NAS 5026ZZNR**	10.5	130	200	208.4	95	94	77	8.5
140	**NAS 5028UUNR**	**NAS 5028ZZNR**	11.0	140	210	218.4	95	94	77	8.5
150	**NAS 5030UUNR**	**NAS 5030ZZNR**	13.5	150	225	233.4	100	99	81	9
160	**NAS 5032UUNR**	**NAS 5032ZZNR**	16.5	160	240	248.4	109	108	89	9.5
170	**NAS 5034UUNR**	**NAS 5034ZZNR**	22.5	170	260	270	122	121	99	11

Notes(1) Minimum allowable value of chamfer dimension r

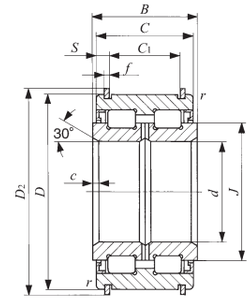
(2) Allowable rotational speed applies to grease lubrication. Considering that the axial load acts under practical operating conditions, up to 1/10 of this value is recommended for actual use.

Remarks1. The inner ring has an oil groove and two oil holes.

2. Roller Bearings for Sheaves are provided with prepacked grease.



NAS50···UUNR



NAS50···ZZNR

f	c	$r_{s\,min}$ (1)	J	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
2	1.5	0.6	50	79 500	116 000	2 500
2	1.5	0.6	56	95 500	144 000	2 000
2	1.5	0.6	61	100 000	158 000	2 000
2.5	2	0.6	68	118 000	193 000	1 800
2.5	2	0.6	73	123 000	208 000	1 700
2.5	2	0.6	78	128 000	224 000	1 600
2.5	2	0.6	84	171 000	284 000	1 400
2.5	2	0.6	91	179 000	308 000	1 300
2.5	2	0.6	97	251 000	428 000	1 300
2.5	2	0.6	101	257 000	446 000	1 200
2.5	2.5	0.6	110	305 000	540 000	1 100
2.5	2.5	0.6	114	312 000	562 000	1 100
2.5	2.5	0.6	118	318 000	584 000	1 000
2.5	3	1	130	384 000	697 000	900
3	3	1	139.5	400 000	750 000	850
3	3	1	156	537 000	1 000 000	750
3	3	1	167	543 000	1 070 000	700
3	3.5	1	176.5	623 000	1 210 000	650
3	3.5	1.5	188.5	720 000	1 390 000	650
4	3.5	1.5	204.5	857 000	1 730 000	600

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

ROLLER BEARINGS

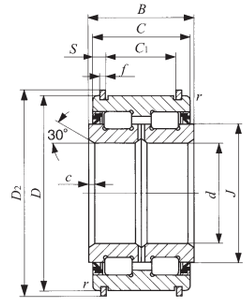
Roller Bearings for Sheaves



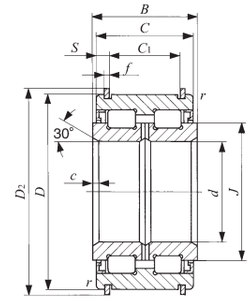


Shaft dia. 180 — 440mm

Shaft dia. mm	Identification number: Sealed type	Identification number: Shield type	Mass (Ref.) kg	Boundary dimensions mm: d	D	D_2	B	C	C_1	S
180	**NAS 5036UUNR**	**NAS 5036ZZNR**	30.0	180	280	294	136	135	110	12.5
190	**NAS 5038UUNR**	**NAS 5038ZZNR**	31.5	190	290	306	136	135	110	12.5
200	**NAS 5040UUNR**	**NAS 5040ZZNR**	40.5	200	310	326	150	149	120	14.5
220	**NAS 5044UUNR**	**NAS 5044ZZNR**	52.0	220	340	356	160	159	130	14.5
240	**NAS 5048UUNR**	**NAS 5048ZZNR**	55.5	240	360	376	160	159	130	14.5
260	**NAS 5052UUNR**	**NAS 5052ZZNR**	85.0	260	400	416	190	189	154	17.5
280	**NAS 5056UUNR**	**NAS 5056ZZNR**	90.9	280	420	440	190	189	154	17.5
300	**NAS 5060UU**	**NAS 5060ZZ**	130	300	460	—	218	216	—	—
320	**NAS 5064UU**	**NAS 5064ZZ**	135	320	480	—	218	216	—	—
340	**NAS 5068UU**	**NAS 5068ZZ**	180	340	520	—	243	241	—	—
360	**NAS 5072UU**	**NAS 5072ZZ**	190	360	540	—	243	241	—	—
380	**NAS 5076UU**	**NAS 5076ZZ**	200	380	560	—	243	241	—	—
400	**NAS 5080UU**	**NAS 5080ZZ**	265	400	600	—	272	270	—	—
420	**NAS 5084UU**	**NAS 5084ZZ**	275	420	620	—	272	270	—	—
440	**NAS 5088UU**	**NAS 5088ZZ**	310	440	650	—	280	278	—	—



NAS50···UUNR



NAS50···ZZNR

f	c	$r_{s\,min}$ (1)	J	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
5	3.5	1.5	217	1 070 000	2 140 000	550
5	3.5	1.5	225	1 120 000	2 230 000	500
5	3.5	1.5	242	1 310 000	2 650 000	500
6	4	1.5	260	1 510 000	3 110 000	450
6	4	1.5	278.5	1 570 000	3 350 000	400
7	5	2	312	2 130 000	4 510 000	350
7	5	2	335	2 210 000	4 860 000	350
—	5	2	359	2 670 000	5 870 000	300
—	5	2	375	2 700 000	6 140 000	300
—	6	2.5	404	3 370 000	7 560 000	300
—	6	2.5	423	3 420 000	7 940 000	250
—	6	2.5	442	3 580 000	8 300 000	250
—	6	2.5	471	4 250 000	10 100 000	250
—	6	2.5	490	4 390 000	10 400 000	250
—	8	3	516	4 570 000	10 900 000	200

Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to grease lubrication. Considering that the axial load acts under practical operating conditions, up to 1/10 of this value is recommended for actual use.

Remarks1. The bearings with a bore diameter d of 300 mm or more has neither stop rings nor stop ring grooves.
2. The inner ring has an oil groove and two oil holes.
3. Roller Bearings for Sheaves are provided with prepacked grease.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch